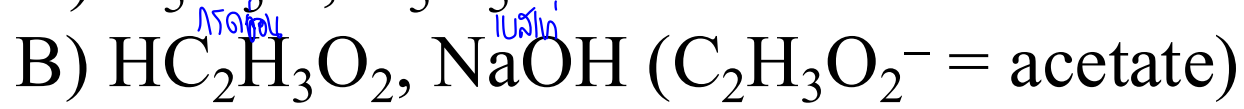


Chapter 17

Additional Aspects of Aqueous Equilibria

1. Which one of the following pairs cannot be mixed together to form a buffer solution? C



Ways to Make a Buffer

- 1) Mix a weak acid and a salt of its conjugate base or a weak base and a salt of its conjugate acid.
- 2) Add strong acid and partially neutralize a weak base or add strong base and partially neutralize a weak acid.

2. The addition of sodium hydroxide and E to water produces a buffer solution.

A) HCl *အဆိပ်*

B) $\text{NaC}_2\text{H}_3\text{O}_2$ *ကဲဆီဗေ*

C) NaF *အဆိပ်*

D) NH_3 *ဗေဒနာ*

E) none of the above

3. Of the following solutions, which has the greatest buffering capacity? A Molar ratio

A) 1.15 M HF and 0.624 M NaF

B) 0.574 M HF and 0.312 M NaF

C) 0.287 M HF and 0.156 M NaF

D) 0.189 M HF and 0.103 M NaF

E) They are all buffer solutions and would all have the same capacity.

4. Which of the following could be added to a solution of sodium acetate to produce a buffer? B

A) acetic acid only

B) acetic acid or hydrochloric acid

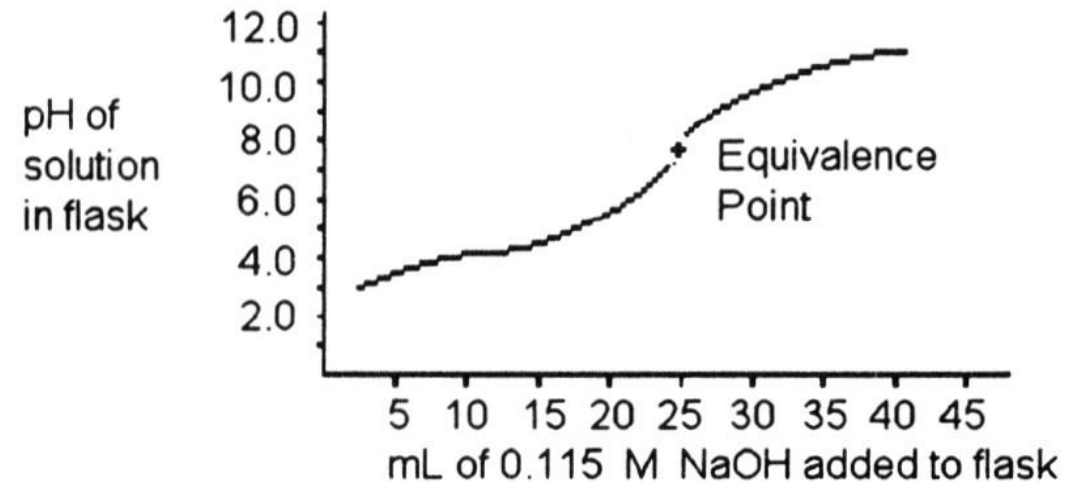
C) hydrochloric acid only

D) potassium acetate only

E) sodium chloride or potassium acetate

5. A 25.0 mL sample of a solution of an unknown compound is titrated with a 0.115 M NaOH solution. The titration curve above was obtained. The unknown compound is C.

- A) a strong acid
- B) a strong base
- ☒ C) a weak acid *Eq. pt above 7*
- D) a weak base
- E) neither an acid nor a base



6. The solubility of lead (II) chloride (PbCl_2) is 1.6×10^{-2} M. What is the K_{sp} of PbCl_2 ? **D**

- A) 5.0×10^{-4}
- B) 4.1×10^{-6}
- C) 3.1×10^{-7}
- D) 1.6×10^{-5}
- E) 1.6×10^{-2}

$$[\text{Cl}^-] = 3.2 \times 10^{-2}$$

$$\therefore K_{sp} = (1.6 \times 10^{-2})(3.2 \times 10^{-2})^2 = 1.63 \times 10^{-5}$$

7. What is the solubility (in M) of PbCl_2 in a 0.15 M solution of HCl? The K_{sp} of PbCl_2 is 1.6×10^{-5} . D

A) 2.0×10^{-3}

B) 1.1×10^{-4}

C) 1.8×10^{-4}

D) 7.1×10^{-4}

E) 1.6×10^{-5}

$$\begin{aligned} 1.6 \times 10^{-5} &= [\text{Pb}^{2+}][\text{Cl}^-]^2 \\ &= [\text{Pb}^{2+}](0.15)^2 \\ [\text{Pb}^{2+}] &= 7.1 \times 10^{-4} \end{aligned}$$

8. Consider a solution containing 0.100 M fluoride ions and 0.126 M hydrogen fluoride. The concentration of hydrogen fluoride after addition of 9.00 mL of 0.0100 M HCl to 25.0 mL of this solution is A M.

$$9 \times 10^{-5} \text{ mol}$$

- A) 0.0709
- B) 0.0900
- C) 0.130
- D) 0.122
- E) 0.00976

$$0.125$$

$$25 \times 10^{-2} \times 1.26 \times 10^{-1}$$

$$\begin{array}{ll} 3.15 \times 10^{-3} \text{ mol } \text{H}^+ & 3.15 \times 10^{-3} \text{ mol } \text{F}^- \\ 0.09 \times 10^{-3} \text{ mol } \text{H}^+ & 2.5 \times 10^{-3} \text{ mol } \text{F}^- \\ \text{(From HF)} & \end{array}$$

10. The pH of a solution prepared by mixing 55.0 mL of 0.183 M KOH and 10.0 mL of 0.145 M HC₂H₃O₂ is D.
0.010065 mol
0.00145

A) 9.97

B) 7.74

C) 0.878

D) 13.122

E) none of the above